



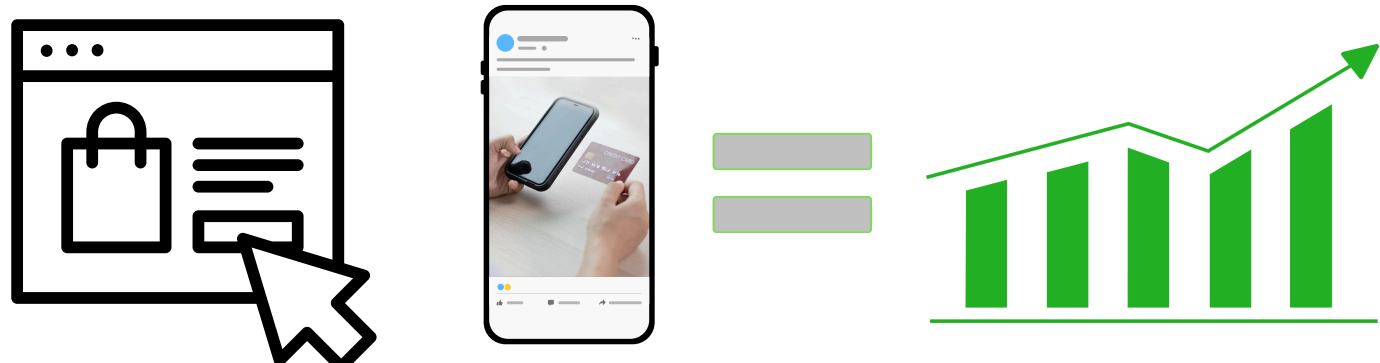
Block Box



DEVELOPING A SECURE LOCKER WITH PARCEL VERIFICATION FOR BLOCKCHAIN INTEGRATION

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1 BACKGROUND



E-commerce growth has led to an increase in peer-to-peer transactions that require secure and reliable last-mile delivery (LMD) solutions. Traditional LMD systems pose challenges related to asymmetric information between buyers and sellers.

Are you sure the contents within this box are as advertised?



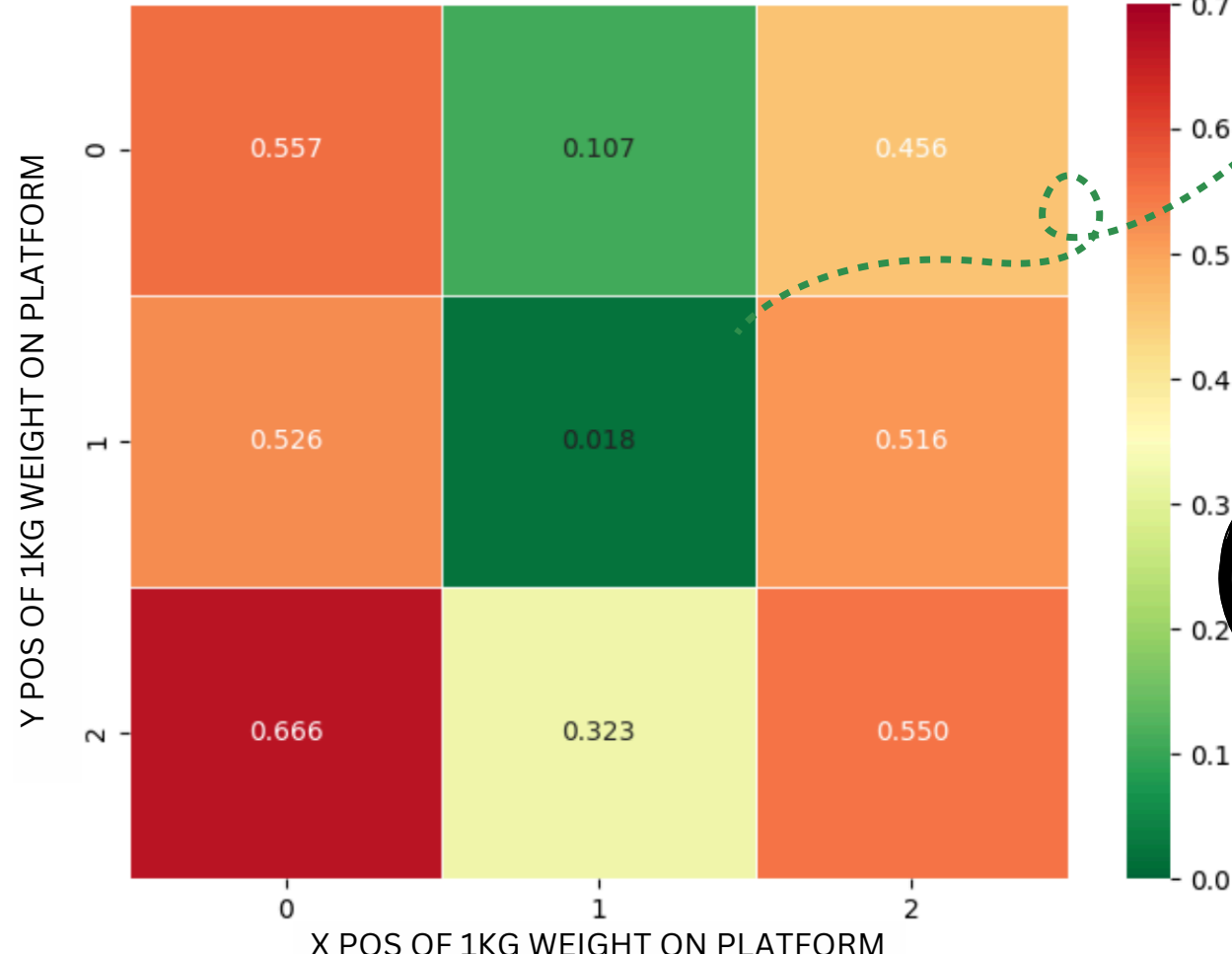
I am absolutely certain.

BUYER

SELLER

4 RESULTS

SENSOR PLATFORM ACCURACY
MEASUREMENT ERROR

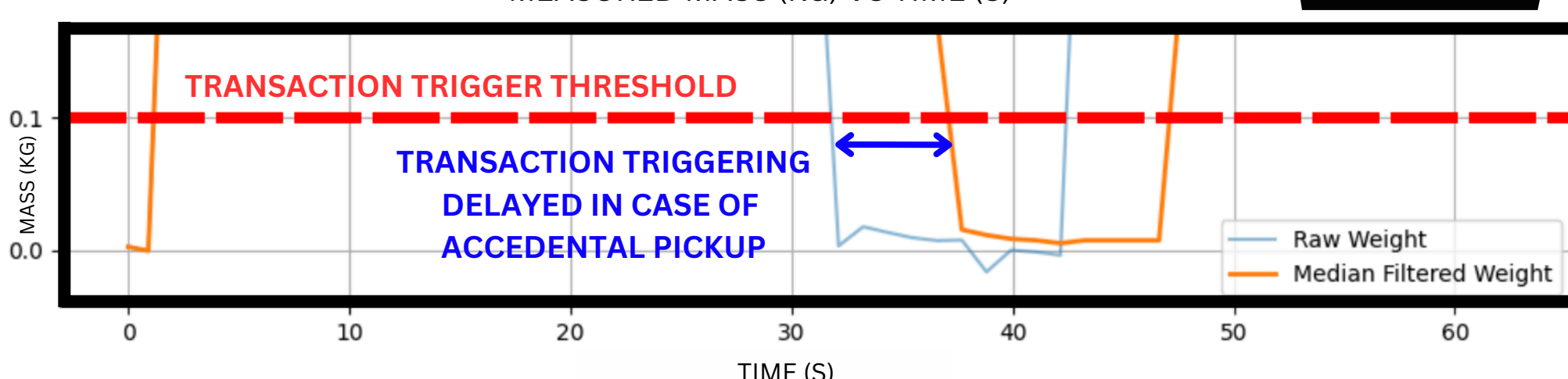


*Scale is the most accurate at the center as off-center error can be quite large as illustrated.

Collection via BLock Box is so efficient because as long as I do not pick my item up for too long while checking it my money does not leave my blockchain wallet.

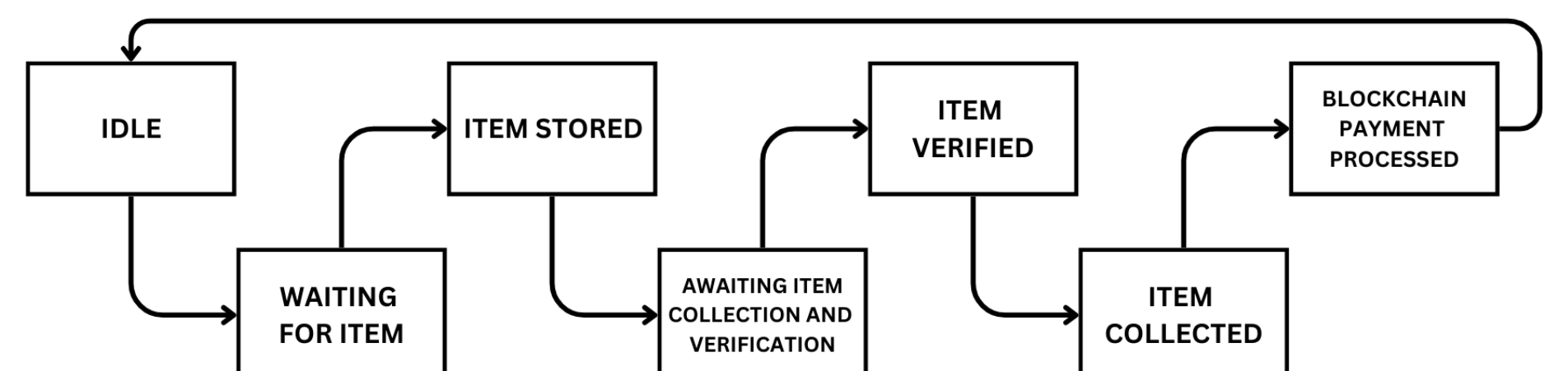
MEDIAN FILTER APPLICATION

MEASURED MASS (KG) VS TIME (S)

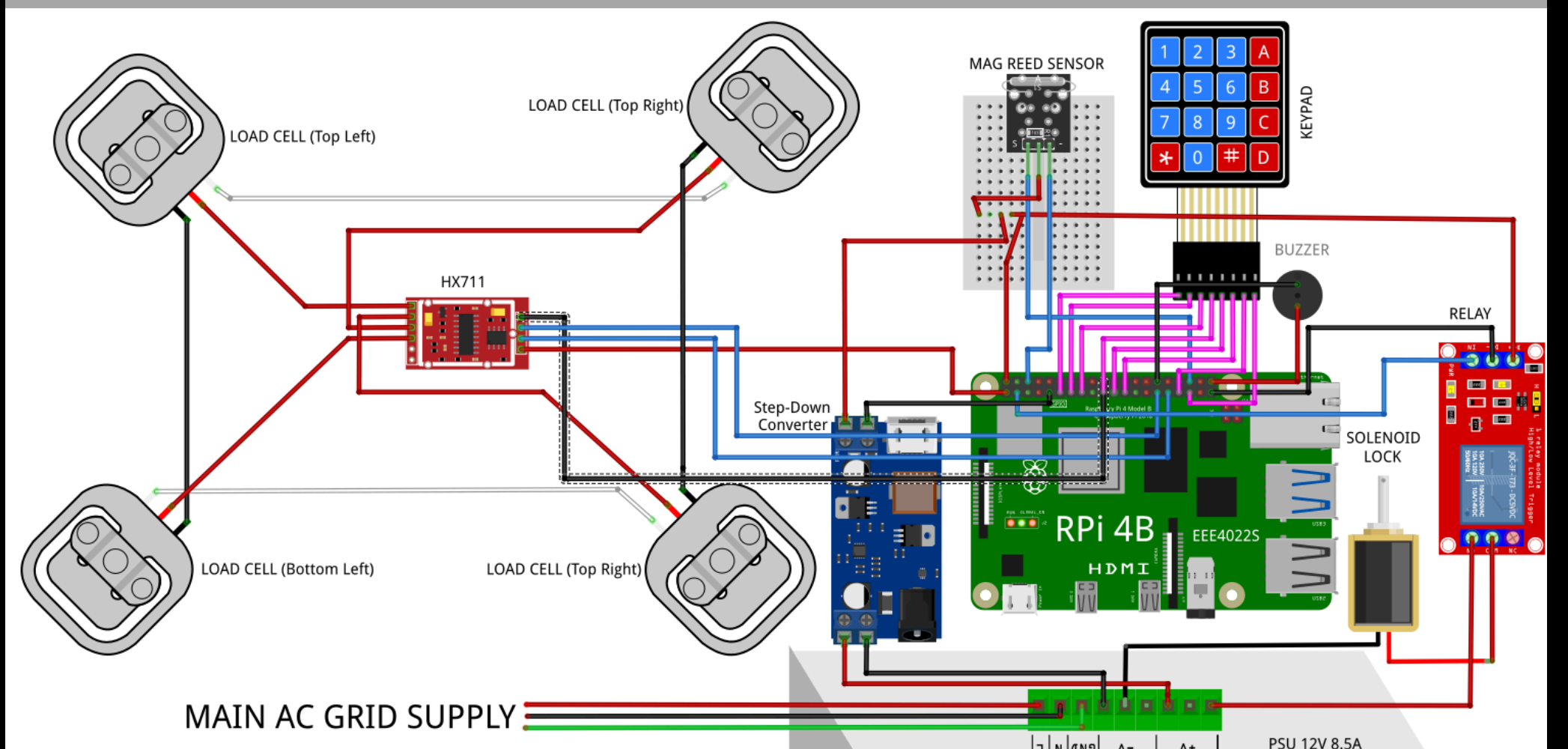


2 APPROACH

The seller typically has more information about the transaction (e.g. condition of the product) when payment takes place. BLock Box solves the problem with the following system states:



3 IMPLEMENTATION



Blockchain Integration

INPUT RESPECTIVE USER DETAILS
TRACK RESPECTIVE USER INTERACTIONS WITH SYSTEM UNDER SYSTEM STATE
SMART CONTRACT PAYMENT TRIGGER ON BUYER ITEM PICK UP
HOST PUBLIC JSON FORMATTED END-POINT OF SYSTEM STATE
CHAINLINK JOB SPEC (EXTERNAL ADAPTERS) DEVELOPMENT WITH HTTPS REQUEST OF JSON ENDPOINT
TRIGGER PAYMENT ON BUYER ITEM PICK UP (USING CHAINLINK)

CURRENT FUNCTIONALITY

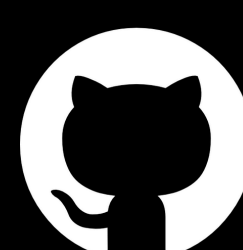
FUTURE DEVELOPMENT

5 CONCLUSION

A secure parcel locker system with automated verification using weight sensors was developed. Upon successful verification, the system uses a smart contract payment system to automate the payment process, therefore, balancing the scale.



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